Limits – 3

Asymptotics of sequences

Homework

1 сентября 2026 года

Theoretical minimum. The lemma on a neutral fixed point $x_{n+1} = x_n - ax_n^p + o(x_n^p)$; the Stolz–Cesàro theorem; telescoping in powers of x_n; comparison of a sum with an integral; the asymptotics $H_n = \ln n + \gamma + O(1/n)$; the step from "the limit is zero" to the leading term together with an estimate of the remainder.

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1. Let $x_0 \in [-1, 1]$ and $x_{n+1} = x_n - \arcsin(\sin^3 x_n)$ for $n \geqslant 0$. Compute $\lim_{n\to\infty} \sqrt{n}\, x_n$.

2. Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous and suppose the sequence $a_n = \int_0^1 f(n + x)\, dx$ converges. Prove that the sequence $b_n = \int_0^1 f(nx)\, dx$ converges as well.

3. (AKHIMO 2018) Let $x_1 = 2018$ and $\frac{x_{n+1}}{x_n} = 2018 + \frac{2018}{n}$ for $n \geqslant 1$. Find $\lim_{n\to\infty} \left(\frac{\log_2 n}{n - \log_{2018} x_n} \right)$.

4. (AKHIMO 2025) Define $b_1 = 1$, $b_2 = 3$, $b_3 = 2$ and, for $n \geqslant 4$, let b_n be the average of the two smaller numbers among $b_{n-1}, b_{n-2}, b_{n-3}$. Find $\lim_{n\to\infty} b_n$.

5. Let $c_0, c_1 > 0$ and $c_{n+1} = \sqrt{c_n} + \sqrt{c_{n-1}}$ for $n \geqslant 1$. Prove that the sequence converges and find its limit.

6. (IMC 2024) Define $x_1 = 2$, $x_2 = 4$ and $x_{n+2} = 3x_{n+1} - 2x_n + \frac{2^n}{x_n}$ for $n \geqslant 1$. Prove that $\lim_{n\to\infty} \frac{x_n}{2^n}$ exists, and estimate it.

7. (OMOUS 2025) Compute $\displaystyle\prod_{n=1}^{\infty} \frac{(2n)^4 - 1}{(2n+1)^4 - 1} \cdot \frac{(n+1)^2}{n^2}$.

8. (Vienna MC 2025) Let (a_n) satisfy $a_0 > 0$ and $a_{n+1} = \frac{a_n}{a_n^2 + a_n + 1}$ for all n. Show that $a_n \to 0$ and determine $\lim_{n\to\infty} n a_n$.

9. For every real x_1 construct the sequence by $x_{n+1} = x_n\big(x_n + \frac{1}{n}\big)$. Prove that there is exactly one value of x_1 for which $0 < x_n < x_{n+1} < 1$ for all n.

10. Let $f\colon (0,\infty) \to \mathbb{R}$ be continuous and suppose that $\lim_{n\to\infty} f(nx) = 0$ for every fixed $x > 0$. Prove that $\lim_{x\to\infty} f(x) = 0$. *(Hint: Baire's category theorem.)*